Lab: The Prediction Error Hunt

Objective

Train your perceptual system to detect prediction errors — moments where reality deviates from expectation — in a real-world setting. Then evaluate whether each prediction error represents a genuine invention opportunity or noise. By the end of this lab, you will have practiced systematic observation, evidence gathering, and signal-noise discrimination.

Materials and Prerequisites

- A notebook or note-taking device
- Access to a real-world environment relevant to your interests (a kitchen, office, workshop, classroom, store, park, etc.)
- 20 minutes of uninterrupted observation time in that environment
- No special tools or technical background required

Part 1: Calibrating Your Perceptual Model (10 minutes)

Before you observe, make your existing expectations explicit. This helps you notice when reality deviates from what you predicted.

Choose an environment to observe. This should be a space you are familiar with and relevant to your invention interests. Examples: your kitchen, your workplace, a classroom, a grocery store, a bus stop, a playground, a craft studio, a gym.

Environment chosen:

Write your response here

Write 10 predictions about how people (including yourself) will use this environment. These should be specific, testable predictions. Examples: "People will enter through the main door," "The dishwasher will be loaded by stacking plates vertically," "Students will take notes on laptops."

#	Prediction	Confidence (1-10)
1		

Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

| | 4 |

Write your response here

|

Write your response here

| | 5 |

Write your response here

|

Write your response here

| | 6 |

Write your response here

|

Write your response here

| | 7 |

Write your response here

|

Write your response here

| | 8 |

Write your response here

|

Write your response here

| | 9 |

Write your response here

|

Write your response here

| | 10 |

Write your response here

|

Write your response here

|

Part 2: Active Observation (20 minutes)

Spend 20 minutes observing your chosen environment. Focus on moments where reality differs from your predictions. Note every deviation, no matter how small.

Rules for observation: - Do not intervene or change anything — just watch - Note what people do, not what they say they do - Pay attention to workarounds, frustrations, repeated actions, abandoned attempts, and improvised solutions - Look at what people do with their hands, faces, and bodies — nonverbal behavior reveals problems that words often hide

Record your observations:

#	What I Predicted	What Actually Happened	Prediction Error (the gap)
1			

Write your response here

|

Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

|

Write your response here

| | 4 |

Write your response here

|

Write your response here

|

Write your response here

| | 5 |

Write your response here

|

Write your response here

|

Write your response here

| | 6 |

Write your response here

|

Write your response here

|

Write your response here

| | 7 |

Write your response here

|

Write your response here

|

Write your response here

| | 8 |

Write your response here

|

Write your response here

|

Write your response here

|

Did you observe any workarounds (improvised solutions people have created for unmet needs)?

Write your response here

Did you notice anything that surprised you — something you had previously never noticed despite visiting this environment regularly?

Write your response here

Part 3: Signal vs. Noise Analysis (15 minutes)

Now evaluate each prediction error to determine whether it represents a genuine invention opportunity (signal) or a transient anomaly (noise).

For each prediction error from Part 2, rate it on the five signal criteria:

Select your top 3 most interesting prediction errors and analyze them:

Prediction Error A:

Description:

Write your response here

Criterion	Rating (1-5)	Evidence
Persistence: Does this happen repeatedly?		

Write your response here

Write your response here

| | Breadth: Do multiple people experience it? |

Write your response here

|

Write your response here

| | Severity: How much cost/pain/time does it cause? |

Write your response here

|

Write your response here

| | Structural cause: Is it caused by design, not behavior? |

Write your response here

|

Write your response here

| | Tractability: Could a different design solve it? |

Write your response here

|

Write your response here

|

Signal or Noise?

Write your response here

Prediction Error B:

Description:

Write your response here

Criterion	Rating (1-5)	Evidence
Persistence: Does this happen repeatedly?		

Write your response here

|

Write your response here

| | Breadth: Do multiple people experience it? |

Write your response here

|

Write your response here

| | Severity: How much cost/pain/time does it cause? |

Write your response here

|

Write your response here

| | Structural cause: Is it caused by design, not behavior? |

Write your response here

|

Write your response here

| | Tractability: Could a different design solve it? |

Write your response here

|

Write your response here

|

Signal or Noise?

Write your response here

Prediction Error C:

Description:

Write your response here

Criterion	**Rating (1-5)**	**Evidence**
Persistence: Does this happen repeatedly?		

Write your response here

|

Write your response here

| | Breadth: Do multiple people experience it? |

Write your response here

|

Write your response here

| | Severity: How much cost/pain/time does it cause? |

Write your response here

|

Write your response here

| | Structural cause: Is it caused by design, not behavior? |

Write your response here

|

Write your response here

| | Tractability: Could a different design solve it? |

Write your response here

|

Write your response here

|

Signal or Noise?

Write your response here

Part 4: Designing an Active Sensing Plan (10 minutes)

For your strongest signal (the prediction error most likely to represent a genuine invention opportunity), design a plan to gather additional evidence.

The prediction error I will investigate further:

Write your response here

What I currently believe about this problem (my hypothesis):

Write your response here

Evidence that would strengthen my belief (confirmatory evidence):

Write your response here

Evidence that would weaken my belief (disconfirmatory evidence):

Write your response here

Specific observations I should make next (active sensing plan):

#	What to observe	Where to observe it	Why this observation is informative
1			

Write your response here

|

Write your response here

|

Write your response here

Write your response here

|

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

|

Write your response here

|

People I should talk to (and why):

#	Who	Why their perspective is informative
1		

Write your response here

|

Write your response here

| | 2 |

Write your response here

|

Write your response here

| | 3 |

Write your response here

|

Write your response here

|

Part 5: Empathic Perception Exercise (5 minutes)

Choose a user who is very different from you — someone whose generative model would produce different prediction errors when encountering your environment.

User persona:

Write your response here

(e.g., "a 78-year-old with limited hand mobility" or "a 5-year-old child" or "someone who speaks a different language" or "someone using a wheelchair")

What prediction errors would this user experience that you did not notice?

Write your response here

How does this perspective change your understanding of the problems in your environment?

Write your response here

Discussion and Debrief

1. **Prediction calibration**: How accurate were your initial predictions? Were you overconfident (many deviations) or well-calibrated (few surprises)? What does this tell you about the sophistication of your generative model for this environment?

2. **Signal density**: How many of your prediction errors passed the signal-noise test? Was it harder to find genuine problems or easier than expected?

3. **Fresh eyes vs. expertise**: Did your familiarity with the environment help you notice problems (expert perception) or hinder you (normalizing dysfunction)? How could you cultivate both expert knowledge and fresh perception?

4. **Active sensing**: How would you prioritize your evidence-gathering plan? What is the single most informative observation you could make, and why?

5. **Empathic perception**: How did adopting a different user's perspective change what you noticed? What does this suggest about the value of user diversity in the invention process?

Write your response here